

Shuai Zhang

Senior Staff ML Engineer & Manager | On-Device Agentic AI | Multimodal Perception | Edge AI

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Summary

Senior engineer and people manager with 8+ years shipping computer-vision products on mobile silicon, now leading **on-device agentic pipeline** development for smart glasses, AI PCs, and IoT. Deep expertise in **real-time segmentation, detection, multimodal perception, VLM/LLM reasoning, and on-device model optimization**, with a proven track record deploying flagship features on Snapdragon and leading cross-functional teams. Ph.D. in Mathematics (UC Irvine); strong foundation in optimization and deep learning theory.

Experience

Senior Staff Engineer & Manager (Machine Learning) — Qualcomm Technologies

11/2024 – Present

- Lead a team building **on-device agentic pipelines** — perception, reasoning, planning, and tool use — deployed on **smart glasses, AI PCs, and agentic IoT systems**.
- Drive end-to-end stack: multimodal perception (segmentation, detection, depth), VLM/LLM reasoning, retrieval/memory, and efficient on-device inference under tight latency and power budgets.
- **ForeSea** (lead researcher): agentic multimodal RAG for long-video understanding with person-centric tracking, retrieval, and VideoLLM reasoning. NeurIPS 2025 Expo demo; under review at ECCV 2026.
- **Leading & Primary Organizer, IEEE LPCVC 2025 & 2026** (IEEE Computer Society feature interview): 77 teams from 25 countries; tracks included **open-vocabulary segmentation, monocular depth estimation**, and robust image classification — core AV perception tasks.
- **Leading & Primary Organizer, CVPR Workshop on Efficient Deep Learning for Computer Vision (ECV) 2025 & 2026**.
- Building Qualcomm's edge-AI developer ecosystem to **3000+ members** through challenges, workshops, webinars, and developer-tooling collaborations.

Staff Engineer (Machine Learning) — Qualcomm Technologies

11/2020 – 11/2024

- Developed **Limitless Segmentation** on Snapdragon 8 Elite: flagship real-time semantic segmentation deployed in the Qualcomm camera pipeline.
- Developed **Segmentation-based Cognitive ISP**: winner of the **2023 Edge AI & Vision Product of the Year Award** (525K+ Google results).
- Productionized SOTA instance segmentation (MobileInst, AAAI 2024); structured pruning via nonconvex regularization (IEEE Access 2021).
- Led full model optimization pipeline: quantization, pruning, low-latency inference on Qualcomm AI Hub / mobile SoCs.

Senior Engineer (Machine Learning) — Qualcomm Technologies

07/2017 – 11/2020

- Developed production **biometric perception** (face detection, anti-spoofing, gaze/attention, user adaptation) and camera segmentation features for on-device deployment.
- Top finishes in low-power vision competitions: 2nd place LPIRC 2018, 3rd place LPIRC 2019.

Data Scientist Intern — Zillow Group

06/2016 – 09/2016

- Recommendation system and house-price estimation (Zestimate pipeline).

Selected Publications (full list: <https://zsivine.github.io/research>)

- **DiSa: Saliency-Aware Foreground-Background Disentangled Framework for Open-Vocabulary Semantic Segmentation.** arXiv:2601.20064, 2026.
- **ForeSea: AI Forensic Search with Multi-modal Queries for Video Surveillance.** arXiv:2603.22872, 2026 — under review at ECCV 2026.
- **Latent Bridge: Feature Delta Prediction for Efficient Dual-System Vision-Language-Action Model Inference.** arXiv:2605.02739, 2026.
- **SEMA: A Scalable and Efficient Mamba-like Attention via Token Localization and Averaging.** ICML 2026.
- **MobileInst: Video Instance Segmentation on the Mobile.** AAAI 2024.

- **AutoShuffleNet: Learning Permutation Matrices via an Exact Lipschitz Continuous Penalty in Deep CNNs.** KDD 2020 — Most Influential KDD Paper.
- **TRP: Trained Rank Pruning for Efficient Deep Neural Networks.** IJCAI 2020.
- **Tiny-Hourglassnet: An Efficient Design for 3D Human Pose Estimation.** IEEE ICIP 2020.
- **Understanding Straight-Through Estimator in Training Activation Quantized Neural Nets.** ICLR 2019.
- **DAC: Data-free Automatic Acceleration of Convolutional Networks.** IEEE WACV 2019 — co-first author.
- **BinaryRelax: A Relaxation Approach for Training DNNs with Quantized Weights.** SIAM Journal on Imaging Sciences, 2018 — co-first author.

Selected Patents

- **S. Zhang,** X. Ying, J. Lyu, Y. Qi. System and method for performing semantic image segmentation (depth-fused cross-attention transformer). *US 12141981* (granted Nov 2024).
- J. Lyu, D. Gao, Y. Qi, **S. Zhang,** N. Bi. Scene segmentation and object tracking. *US 12236614* (granted Feb 2025).
- E. Mequanint, **S. Zhang,** Y. Qi, N. Bi. User adaptation for biometric authentication. *US 11887404* (granted Jan 2024).
- E. Mequanint, **S. Zhang,** Y. Qi, N. Bi. Personalized eye openness estimation. *US App. 16/239,352* (pending).

Awards & Recognition

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| • IEEE Senior Member | 2023 |
| • Edge AI & Vision Product of the Year — Segmentation-based Cognitive ISP (Qualcomm) | 2023 |
| • 3rd Place, Low Latency Object Detection — IEEE LPIRC | 2019 |
| • 2nd Place, Image Classification & Detection — IEEE LPIRC | 2018 |
| • Kovalevsky Outstanding Ph.D. Thesis Award — UC Irvine | 2017 |
| • SIAM Student Travel Award | 2016 |
| • Von Neumann Outstanding Research Award — UC Irvine | 2015 |

Education

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| University of California, Irvine | Irvine, CA |
| • <i>Ph.D., Mathematics</i> | 2017 |
| Advisor: Prof. Jack Xin Research: Compressed Sensing, Image Processing, Machine Learning | |
| • <i>M.S., Mathematics</i> | 2014 |
| Shandong University | Jinan, China |
| • <i>M.S., Computational Mathematics</i> (GPA 3.7/4.0) | 2012 |
| • <i>B.S., Applied Mathematics</i> (GPA 3.83/4.0) | 2009 |